

lab-4-Yue Chenghao

code

```
import java.util.ArrayList;
import java.util.Date;

public class testclass {
    public static void main(String[] args) {
        // Create an ArrayList that can hold objects of any type
        ArrayList<Object> list = new ArrayList<>();

        // Create and add a Loan object
        Loan loan = new Loan(10000, 5, 10);
        list.add(loan);

        // Create and add a Date object (current date and time)
        Date date = new Date();
        list.add(date);

        // Create and add a Circle object
        Circle circle = new Circle(2);
        list.add(circle);

        // Loop through the list and print each object's toString() result
        for (Object obj : list) {
            System.out.println(obj.toString()); // Implicitly calls toString()
        }
    }

    // A simple class representing a Loan
    class Loan {
        private double amount;
        private double interestRate;
        private int years;

        // Constructor
        public Loan(double amount, double interestRate, int years) {
            this.amount = amount;
            this.interestRate = interestRate;
            this.years = years;
        }

        // Override toString() to provide meaningful info
        @Override
```

```

    public String toString() {
        return "Loan: amount = " + amount + ", interest rate = " + interestRate
+ "%, years = " + years;
    }
}

// A simple class representing a Circle
class Circle {
    private double radius;

    // Constructor
    public Circle(double radius) {
        this.radius = radius;
    }

    // Override toString() to display radius and area
    @Override
    public String toString() {
        return "Circle: radius = " + radius + ", area = " + Math.PI * radius *
radius;
    }
}

```

output

```

> cd "/Users/v/Desktop/sem-2/CCS3101/LAB/lab4/" && javac testclass.java && java testclass
Loan: amount = 10000.0, interest rate = 5.0%, years = 10
Thu May 01 23:20:18 GMT+08:00 2025
Circle: radius = 2.0, area = 12.566370614359172

```

```

~/Desktop/sem-2/CCS3101/LAB/lab4 | main !18 ?12
> 

```

Java: Ready 行 6